

SRIE

Project 3 - An AI tutor to prevent cognitive surrender

BILEL HATMI, SUPERVISION

EIGHT WEEKS

4 August to 28
September 2026



Cognitive surrender erodes both the know-how and the judgment that would have caught the error

The case

Four endoscopy centres adopted AI. **Adenoma detection fell six points, on a skill built over years.** Detection rates stood at:

- 28.4% before AI
- 22.4% after routine adoption

The stakes are high: a missed polyp can mean an undiagnosed cancer.¹

Automation erodes skill entirely

- Where the machine runs the routine, practice disappears: **the skill goes blunt**²
- **Recovery is partial** even when the skill is retained (cf. NASA F-1)³



Cognitive surrender, named

- **Cognitive surrender**: any task with cognitive friction is handed to the tool⁴
- Beyond skill, **a metacognitive decline**: a reduced ability to learn⁴



Trust in tools, doubt in oneself

- Despite the human cost, trust **shifts to the tools that deliver productivity**⁴
- Self-perceived skill, weighed against trust in the tool, predicts the surrender

Tools that deliver productivity earn trust, and trust drives adoption. What they erode along the way is **human skill and metacognition**

The next page traces what that loss costs

The loss falls on the young now, a national economy later

The freeze runs through hiring

- Employment for young workers in AI-exposed roles sits 19% below their peers — hiring has frozen¹
- Once hired, they lean on the tool without guardrails: **productivity now**, at the cost of never building the skill

An illusion of mastery, a measured collapse

- Students with AI solved 48% more, then scored 17% worse closed-book²
- The same gap now hits those with nothing built yet²



A national capacity losing its resilience

Innovation needs more minds as creativity per mind falls³

Shared algorithms cut cognitive diversity — **errors correlate even if they're better**⁴

Individual productivity rises while **collective diversity falls**⁵

**Eroded judgment and
cognitive diversity
destroy resilience**

In the short term the damage never shows up in productivity; by the time it does, rebuilding costs far more

The tool of the problem is the tool of the remedy

Bastani ran a second arm of the same experiment, and the cost disappeared: the results suggest AI can serve as a tool for learning

The reversal: scaffolding, not technology

- With a trained AI supervisor, training performance rose 127%; exams matched control¹



A theoretical remedy already existed

- One teacher can only reach thirty students; **a tool scales without limit**
- We only need a tool that represents **a student's learning state**, driven by an LLM. Such a tool was already validated in 1985² 🔍



Knowledge space (KS)

A domain's items and the order in which they must be learned, from which every state a learner can occupy is derived.

Automation removes the cost, making this tool applicable in real settings

- Elicitation can run to a million queries, especially when building the KS, and marking costs hours — the tool does both in seconds
- AI also acts as an instrument, and may prove more precise on the KS, since it only has to measure specific, factual parameters

The obstacle was never the theory but **the cost of building it** — and that is what the LLM erases.

The next page sets out the four questions this project raises.

The project's four open questions

Four questions lead the project: modeling, construction, measurement, strategy.

Modeling the structure

- **Currently incomplete:** the domain's classical structure must be extended **to fold in the metacognitive part**
- **Genuinely open:** what belongs to the state, and how each part is identified, is still unanswered

Automating the construction

- **Whether automatic domain construction is feasible**, judged against what an expert would produce. The elicitation runs unchanged, with the model in the expert's place, on domains whose prerequisite order can be checked by hand

Measuring a state

- Whether the LLM can read a state well: **fitting parameters and handling uncertainty**
- With no real students on this timeline, **LLM personas** will stand in
- **Success: recovering the state from answers and confidence alone**

Adaptive strategy

- Strategies can scale **from local to regional**
- **Not what pleases the user**, but what the classical rule favors in information gain, within a Bayesian framework

Project's worth reaches beyond education

Personalisation is the near case of a general problem: wherever a formal object is too costly for a human to steer, a language model can become the hand that works it with the structure saying where to look (giving the construct), and the model carrying the language (being its instrument and its pilot).

Little capital, a real contribution

- Education earns no category, **1–2% of AI venture capital**¹
- Despite the productivity loss, this offers **a nice compromise between human cognition and time.**

Nobody scaled the scaffolding

- One tutor, not based on a valid structure, just through prompting engineering **raised training performance 127%**, never scaled²

Applicable to other fields once the mathematical structure valid

- Can be **applicable to mental health, introspection...** and so many pain points that are experienced individually in which a cognitive effort is involved

With an LLM, these tools become steerable, and the queries run efficiently.



The supervision

This program is meant to be more than just a research project

More than a project

Four things get built alongside the research, and they are on the timetable like everything else.

Method



- **3 cross-cutting themes**, everyone in the room regardless of project: **literature review, reading and writing a paper, designing an experiment.**
- Turning a qualitative question into **a valid statistical design**, and **criticising an experiment while reading it.**

Tools



- **Claude Code** across the whole range, **from literature to code, wired into a single connected ecosystem.**
- Using **organisational tools like Jira, Drive and Slack** to improve efficiency and communication

Communication



- **Presenting technical work** in a structured way
- **Asking the right questions** at the right moment **to encourage creativity**, and reacting appropriately when knowledge is missing.

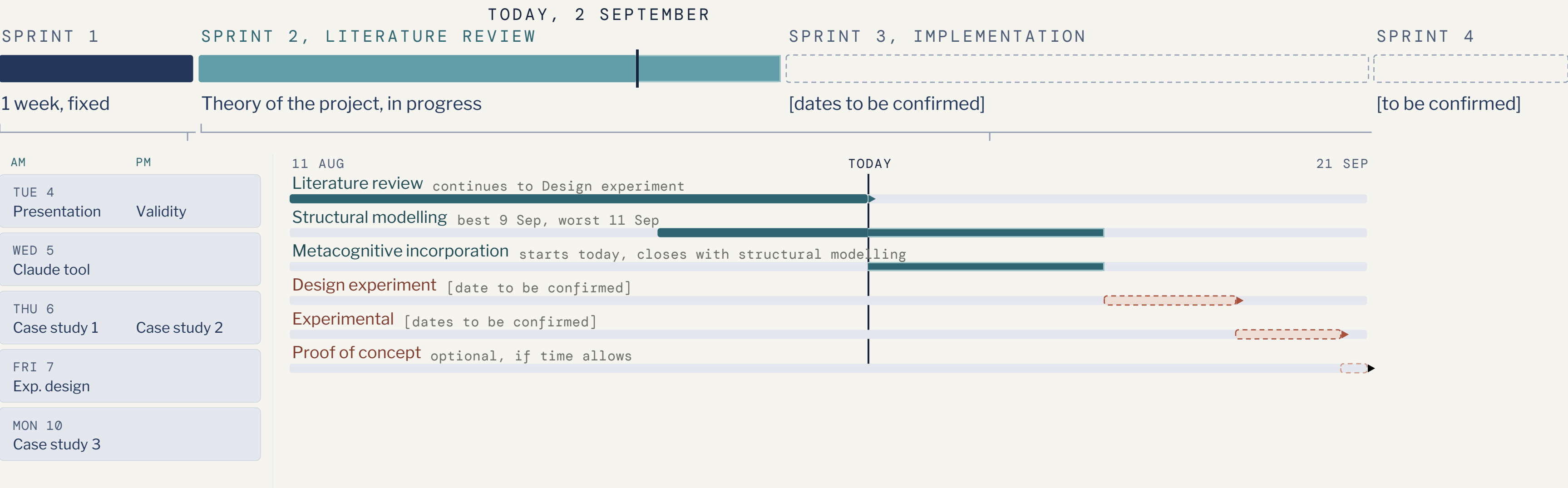
Yourself



- **A time to get to know yourself and work more efficiently** (Productivity, Rest, Learning, feedback culture, fear of comparison, perfectionism...). These are pain points common to highly demanding work such as research.

The team is measuring what AI does to the people who use it, so it is unusually well placed **to study its own use of AI tools.**
The next page sets out the eight-week calendar.

The roadmap



The experimental scope might be reduced, the theoretical phasis being more complex than anticipated

How the stream runs

Three levels, each carrying a different register of work. No level does the job of another.



The whole cohort for **what is deemed useful to everyone**, the group meeting to **keep the needs and ambitions of each stream coherent**, the individual meeting **to delve deeper into technical issues**.

Core bibliography

1. The construct and its construction

Spaces for the Assessment of Knowledge.
Doignon and Falmagne, 1985. International Journal of Man-Machine Studies, 23(2)

How to Build a Knowledge Space by Querying an Expert.
Koppen and Doignon, 1990. Journal of Mathematical Psychology, 34(3)

2. LLMs as cognitive instruments

Using Large Language Models to Assign Partial Credit to Students' Explanations of Problem-Solving Process.
Chen and Wan, 2024. arXiv:2412.06910

How Consistent Is the Algorithm? Examining the Intra- and Inter-Rater Reliability of LLM-Based Writing Assessment.
AlAmir, 2026. Frontiers in Education, 11

3. The adaptative strategy

Faster Teaching via POMDP Planning.
Rafferty et al., 2016. Cognitive Science, 40(6)

Making Things Hard on Yourself, but in a Good Way: Creating Desirable Difficulties to Enhance Learning.
Bjork and Bjork, 2011. In Psychology and the Real World. Worth Publishers

4. Metacognitive vigilance, and evidence of decline

How to Measure Metacognition.
Fleming and Lau, 2014. Frontiers in Human Neuroscience, 8

Generative AI Without Guardrails Can Harm Learning: Evidence from High School Mathematics.
Bastani et al., 2025. PNAS, 122(26)